

Model Predictive Control of Nonholonomic Mobile Robots with Backward Motion

Di Cui * Huiping Li *

** School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an, CO 710072 China (e-mail: dicui, lihuiping@nwpu.edu.cn).*

Abstract: Currently, most of model predictive control (MPC) of mobile robots are designed based on forward movements, which is not efficient when the robot in certain initial positions. In this paper, a novel model predictive controller is designed to solve the regulation problem of a nonholonomic wheeled mobile robot with backward motion when its initial position in the first and second quadrants of Cartesian coordinates. System state variables selection and corresponding kinematic models in polar coordinate are defined to characterize backward movements. The terminal state cost function and terminal region together with the local controller are designed to guarantee the stability of the optimization problem (OP). Comparison studies show that the proposed MPC algorithm outperforms conventional ones.

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Keywords: Model predictive control (MPC), backward motion, nonholonomic wheeled mobile robots, regulation, polar coordinate.

1. INTRODUCTION

The use of mobile robots in various fields has been greatly appreciated, but further increases demands on the motion control of robots. As one of the basic control tasks, point stabilization or regulation of the wheeled mobile robots have attracted considerable attention. It is well known that the wheeled robots is of nonholonomic constraints, and does not meet the necessary conditions proposed by Brockett (1983), which means it is impossible to find a smooth time-invariant stabilizing feedback control law. Several kinds of controllers have been proposed to deal with this problem, e.g., discontinuous time-invariant feedback stabilization control (Sontag (2011); Reyhanoglu (1995); Marchand and Alamir (2003); Fang and Wang (2011)), (smooth) time-varying feedback control (Sordalen and Egeland (1995); Godhavn and Egeland (1997)), hybrid control Panagou and Kyriakopoulos (2014). These methods share a common disadvantage that they do not take into account the constraints of control inputs. However, there widely exist control constraints for mobile robots in practice, e.g. the motor torques to drive the robots can be not too large due to saturation and power limits.

It is well known that model predictive control (MPC) can handle system constraints explicitly by solving constrained optimization problem (OP). The main idea of MPC is to solve OP with respect to predicted control sequence in a finite horizon online based on the current state at each moment, then apply the first element of the optimal control sequence to the system. In this framework, several methods for guaranteeing stability have been proposed in the existing literature (Mayne et al. (2000); Grne and Pannek (2011); Rawlings and Mayne (2009); Chen and Allgöwer (1998)). First, the infinite prediction horizon can easily drive the stability of the closed-loop system to en-

sure stability, but it is generally unrealizable. Keerthi and Gilbert (1988) proves the stability by adding a terminal equality constraint, which means both the terminal state and local control law equal to zero. However, this method may be computationally intensive and waste time when solving QP. Jadbabaie et al. (2001) solves the problem by adopting a terminal cost and corresponding terminal inequality constraint, and this method significantly improves attraction region, but a local linear feedback controller is required. However, such a local control law does not exist for stabilizing the wheeled mobile robots under Cartesian coordinates.

Progress has been made in the regulation of nonholonomic wheeled mobile robots by using MPC. Gu and Hu (2005) uses a Lyapunov-like function and a terminal region to prove the stability, but the algorithm is established only when the wheeled mobile robots moving forward. Xie and Fierro (2008) provides a contractive constraint to the first prediction state, which makes the selection of the initial state is conservative. The state trajectories generated by the above two algorithms eventually reach the origin from the negative part of x axis and have a small curvature near the origin for initial positions of robots in the first and second quadrants, that means the aforementioned algorithms are not applicable for the situation that the curvature of mobile robots is limited. Worthmann et al. (2014) and Worthmann et al. (2015) use suitable cost function and prediction horizon to regulate the wheeled mobile robots, but the stabilizing horizon need to be calculated in advance and the state trajectory is more complex than those simulation results in Gu and Hu (2005) and Xie and Fierro (2008). Deng and Li (2015) provides an regulation application of mobile robots without stability guarantees.

By identifying the drawbacks of Gu and Hu (2005) and the fact that almost all the industrial wheeled mobile robots can move both forward and backward, we study the model predictive regulation problem of mobile robots with backward motion. In this paper, we aim to improving the control performance in Gu and Hu (2005) by designing novel model predictive control algorithm to general backward motion. The main contribution of the work can be two-folds:

- We established the kinematic model for backward movement of mobile robots, and provided an optional choice for the initial position in the first and second quadrants to handle the problem of curvature limitation near the origin.
- We designed the novel model predictive control algorithm with corresponding terminal region and local control law, and proved that the closed-loop system is asymptotically stability in backward motion.

The structure of the paper is organized as follows: The detailed kinematic model and regulation problem of non-holonomic wheeled mobile robots for backward movement are specified in Section 2. In Section 3, the feasibility and stability of the closed-loop system are proved, then we present the design of the terminal region together with local control law. In Section 4, simulation and comparison studies are shown. Finally, conclusions are briefly drawn in Section 5.

2. PROBLEM FORMULATION

2.1 Kinematic Model

In this subsection, for the ease of presentation, we write variable " $q(t)$ " as " q ". Considering a nonholonomic wheeled mobile robot modeled as follows:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} \cos \psi & 0 \\ \sin \psi & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}, \quad (1)$$

where the position of the robot is denoted as $(x, y)^T$ and ψ is the orientation of the robot; input v and w are the linear and angular velocity, respectively. Here, we handle the control problem based on the polar coordinate. For the case of robot with forward motion (see Fig. 1(a)), state definition and kinematic model in polar coordinate have been detailed by Gu and Hu (2005). For the case of robot with backward motion (see Figure 1(b)), system state variables $\mathbf{x} \triangleq [e \ \theta \ \alpha]^T$ can be defined as:

$$\begin{aligned} e &= \sqrt{x^2 + y^2}, \\ \theta &= \text{ACTAN}(y, x), \\ \alpha &= \psi - \theta, \end{aligned} \quad (2)$$

where e is the distance from the robot to the origin and θ denotes the four quadrant inverse tangent function represented by

$$\begin{aligned} \theta &= \text{ACTAN}(y, x) \\ &= \begin{cases} \arctan \frac{y}{x} & x > 0, \\ \frac{\pi}{2} & x = 0, y > 0, \\ \pi + \arctan \frac{y}{x} & x < 0, y \geq 0, \\ -\frac{\pi}{2} & x = 0, y < 0, \\ -\pi + \arctan \frac{y}{x} & x < 0, y < 0. \end{cases} \end{aligned} \quad (3)$$

Then the system kinematic model is

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} v \cos \alpha \\ v \sin \alpha \\ w - \frac{e}{v \sin \alpha} \end{bmatrix}, \quad (4)$$

where $\mathbf{u} \triangleq [v \ w]^T$. The state and input constraints are $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^3$ and $\mathbf{u} \in \mathcal{U} \subseteq \mathbb{R}^2$, respectively. Different from forward motion, the constraints of control input are $\bar{v} \leq v \leq 0$ and $|w| \leq \bar{w}$ for the backward case, where $\bar{v} < 0$ and $\bar{w} > 0$ are the minimum linear and maximum angular velocities, respectively.

It should be noticed that in the framework of forward motion, the variables θ and α are defined as $\text{ACTAN}(-y, -x)$ and $\psi - \theta$, respectively. Then the derived system dynamic is not the same as the backward motion.

2.2 Regulation Controller

At each time instance t_k , the system control input can be derived by solving an optimal control problem $\mathbf{P}$, which is formulated as follows:

$$\begin{aligned} \min_{\mathbf{u}^*(s; t_k)} & J(\bar{\mathbf{u}}(s; t_k); t_k, \bar{\mathbf{x}}(t_k; t_k)) \\ \text{s.t.} & \dot{\bar{\mathbf{x}}}(s; t_k) = f(\bar{\mathbf{x}}(s; t_k), \bar{\mathbf{u}}(s; t_k)), \\ & \bar{\mathbf{x}}(s; t_k) \in \mathcal{X}, \\ & \bar{\mathbf{u}}(s; t_k) \in \mathcal{U}, \\ & \bar{\mathbf{x}}(t_k + T_p; t_k) \in \Omega, \end{aligned} \quad (5)$$

where $s \in [t_k, t_k + T_p]$; Ω is the terminal region; $\bar{\mathbf{x}}(t_k; t_k)$ denote the initial system state $\mathbf{x}(t_k)$ and T_p is the prediction horizon. The detailed expression of the cost function J is:

$$\begin{aligned} J &\triangleq \int_{t_k}^{t_k + T_p} [\|\bar{\mathbf{x}}(s; t_k)\|_Q^2 + \|\bar{\mathbf{u}}(s; t_k)\|_R^2] ds \\ &+ \|\bar{\mathbf{x}}(t_k + T_p; t_k)\|_P^2. \end{aligned} \quad (6)$$

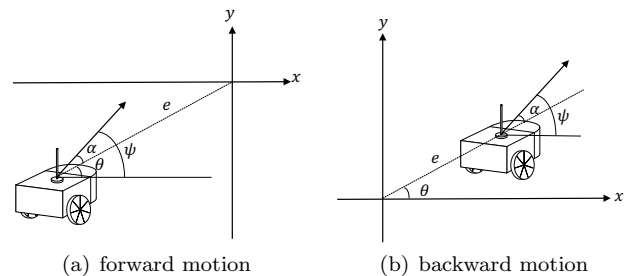


Fig. 1. state definition of mobile robots for the case of (a) forward motion (b) backward motion

Here, both $Q = \text{diag}(q_1, q_2, q_3)$ and $R = \text{diag}(r_1, r_2)$ are positive definite matrices. $\|\bar{\mathbf{x}}(t_k + T_p; t_k)\|_P^2$ is the terminal cost which will be specified in Section 3.

For the convenience of understanding, $\bar{\mathbf{u}}(\cdot)$ and $\mathbf{u}^*(\cdot)$ describe the feasible and the optimal control input, respectively. The related system states are defined accordingly.

At time instant t_k , the open-loop optimal control input is denoted by $\mathbf{u}^*(s, t_k)$, where $s \in [t_k, t_k + T_p]$. The procedures of MPC are as follows: At each time instant t_k , the controller solves the problem $\mathbf{P}$ in (5) to obtain the optimal control sequence $\mathbf{u}^*(s, t_k)$; by applying the first part (the first sampling interval δ) of the optimal control input to the system, we get the closed-loop system input $\mathbf{u}^*(t) = \mathbf{u}^*(s, t_k)$ for $t \in [t_k, t_k + \delta]$; let $t_{k+1} = t_k + \delta$ and repeat the aforementioned steps. Finally, the closed-loop system given by the MPC algorithm is:

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(0), \mathbf{u}^*(t)). \quad (7)$$

3. FEASIBILITY AND STABILITY ANALYSIS

3.1 Feasibility and Stability

In this section, we analyze the feasibility of the algorithm and the closed-loop stability. Defining $F(\mathbf{x}) = \|\mathbf{x}\|_P^2$.

Assumption 1. There exists a control input $\bar{\mathbf{u}}(s, t_0) \in \mathcal{U}$, $s \in [t_0, t_0 + T_p]$ such that the state trajectory generated by (4) with initial state $\mathbf{x}(t_0)$ meets the state constraints in $\mathbf{P}$.

Assumption 2. There exists a local control law $\mathbf{u} = K\mathbf{x} \in \mathcal{U}$ such that for the system (4), we have:

$$\dot{F}(\mathbf{x}) \leq -\|\mathbf{x}\|_{Q^*}^2, \forall \mathbf{x} \in \Omega, \quad (8)$$

where $Q^* = Q + K^T R K$.

Assumption 2 shows that the terminal cost $F(\mathbf{x})$ is a local lyapunov function. In fact, it is a sufficient condition to guarantee the system stability, which will be detailed in the proof of theorem 2.

Theorem 1. For regulation problem of the wheeled mobile robots, if Assumption 1 and 2 satisfy, the designed MPC algorithm is feasible all the time.

Proof. It is supposed that there exists an optimal control input $\mathbf{u}^*(s, t_k)$, $s \in [t_k, t_k + T_p]$ to the problem $\mathbf{P}$ at time instance t_k with initial system state $\mathbf{x}(t_k)$, then the corresponding system trajectory derived by (4) can be denoted as $\mathbf{x}^*(s, t_k)$. At next sampling instance t_{k+1} , the initial system state is $\mathbf{x}(t_{k+1}) = \mathbf{x}^*(t_{k+1}, t_k)$. Constructing a feasible system input based on Assumption 2 as Chen and Allgöwer (1998):

$$\bar{\mathbf{u}}(s, t_{k+1}) = \begin{cases} \mathbf{u}^*(s, t_k) & s \in [t_k, t_k + T_p], \\ K\bar{\mathbf{x}}(s, t_k + T_p) & s \in [t_k + T_p, t_{k+1} + T_p]. \end{cases} \quad (9)$$

where $\bar{\mathbf{x}}(s, t_k + T_p)$ is the system trajectory generated by (4) with $\bar{\mathbf{x}}(t_k + T_p, t_k + T_p) = \mathbf{x}^*(t_k + T_p, t_k)$. By following the procedures in Chen and Allgöwer (1998), the recursive feasibility can easily be guaranteed by Assumption 1.

Theorem 2. The closed-loop system (7) is asymptotically stable if Assumption 2 holds.

Proof. First, due to the optimality of the cost function, we have:

$$\begin{aligned} & J^*(\mathbf{u}^*(s, t_{k+1}), t_{k+1}, \mathbf{x}(t_{k+1})) - J^*(\mathbf{u}^*(s, t_k), t_k, \mathbf{x}(t_k)) \\ & \leq \bar{J}(\bar{\mathbf{u}}(s, t_{k+1}), t_{k+1}, \mathbf{x}(t_{k+1})) - J^*(\mathbf{u}^*(s, t_k), t_k, \mathbf{x}(t_k)). \end{aligned} \quad (10)$$

Then we prove the monotonically decreasing property of the optimal cost function. By selecting $\bar{\mathbf{u}}(s, t_{k+1})$ as (9), we get:

$$\begin{aligned} & \bar{J}(\bar{\mathbf{u}}(s, t_{k+1}), t_{k+1}, \mathbf{x}(t_{k+1})) - J^*(\mathbf{u}^*(s, t_k), t_k, \mathbf{x}(t_k)) \\ & \leq \int_{t_k + T_p}^{t_{k+1} + T_p} \|\bar{\mathbf{x}}(s; t_k + T_p)\|_{Q^*}^2 ds \\ & \quad - \int_{t_k}^{t_{k+1}} \|\mathbf{x}^*(s; t_k)\|_Q^2 + \|\mathbf{u}^*(s; t_k)\|_R^2 ds \\ & \quad + \|\bar{\mathbf{x}}(t_{k+1} + T_p; t_k)\|_P^2 - \|\mathbf{x}^*(t_k + T_p; t_k)\|_P^2. \end{aligned} \quad (11)$$

Here, by integrating both side of (8) 2 from $t_k + T_p$ to $t_{k+1} + T_p$ with $\mathbf{x} = \bar{\mathbf{x}}(s; t_k + T_p)$, it has:

$$\begin{aligned} & \int_{t_k + T_p}^{t_{k+1} + T_p} \|\bar{\mathbf{x}}(s; t_k + T_p)\|_{Q^*}^2 ds \\ & \leq \|\mathbf{x}^*(t_k + T_p; t_k)\|_P^2 - \|\bar{\mathbf{x}}(t_{k+1} + T_p; t_k)\|_P^2. \end{aligned} \quad (12)$$

Substitute (12) into the inequality (11), we finally yields:

$$\begin{aligned} & \bar{J}(\bar{\mathbf{u}}(s, t_{k+1}), t_{k+1}, \mathbf{x}(t_{k+1})) - J^*(\mathbf{u}^*(s, t_k), t_k, \mathbf{x}(t_k)) \\ & \leq - \int_{t_k}^{t_{k+1}} \|\mathbf{x}^*(s; t_k)\|_Q^2 + \|\mathbf{u}^*(s; t_k)\|_R^2 ds. \end{aligned} \quad (13)$$

Motivated by Chen and Allgöwer (1998), The proof of the asymptotically stability can be completed with the combination of (10) and (13).

3.2 Terminal Region and Local Control Law

In this subsection, we give the detailed formulation of the terminal cost $F(\mathbf{x})$ and then derive the terminal region and local control law to satisfy Assumption 2.

Here, the terminal cost function in (8) is designed as:

$$F(\mathbf{x}) = \frac{1}{2}e^2 + \frac{1}{2}(\alpha + \theta)^2. \quad (14)$$

According to the Assumption A4 in Mayne et al. (2000), $F(\mathbf{x})$ is required to be a local Lyapunov function, which means $F(\mathbf{0}) = 0$ and $F(\mathbf{x}) > 0, \mathbf{x} \neq \mathbf{0}$. Thus, we have:

$$\alpha\theta \geq 0. \quad (15)$$

Choosing the local control law (linear and angular velocity) as:

$$\begin{aligned} v &= -\gamma e & \gamma > 0, \\ w &= -\xi(\theta + \alpha) & \xi > 0, \end{aligned} \quad (16)$$

which means

$$K = - \begin{bmatrix} \xi & 0 & 0 \\ 0 & \gamma & \gamma \end{bmatrix} \quad (17)$$

in (8). Considering the system dynamic in (4), $\dot{F}(\mathbf{x}) + \|\mathbf{x}\|_{Q^*}^2$ can be expanded as:

$$\begin{aligned}
& \dot{F}(\mathbf{x}) + \|\mathbf{x}\|_{Q^*}^2 \\
&= e(-\gamma e \cos \alpha) - \xi(\theta + \alpha)^2 + q_1 e^2 + q_2 \theta^2 + q_3 \alpha^2 \\
&\quad + r_1(\gamma e)^2 + r_2 \xi^2 (\theta + \alpha)^2 \\
&= (q_1 + r_1 \gamma^2 - \gamma \cos \alpha) e^2 + (r_2 \xi^2 - \xi + q_2) \theta^2 \\
&\quad + (r_2 \xi^2 - \xi + q_3) \alpha^2 + 2(r_2 \xi^2 - \xi) \alpha \theta.
\end{aligned} \tag{18}$$

Let the coefficients satisfy:

$$\begin{aligned}
r_2 \xi^2 - \xi + q_2 &\geq 0, \\
r_2 \xi^2 - \xi + q_3 &\geq 0, \\
r_2 \xi - 1 &\geq 0
\end{aligned} \tag{19}$$

and define the following constraints for system state α :

$$-\arccos \frac{q_1 + r_1 \gamma^2}{\gamma} \leq \alpha \leq \arccos \frac{q_1 + r_1 \gamma^2}{\gamma}. \tag{20}$$

We finally have $\dot{F}(\mathbf{x}) + \|\mathbf{x}\|_{Q^*}^2 \leq 0$ by substituting (19) (20) into (18). In addition, the local control law need to be constrained by $\mathcal{U}$, which indicates that:

$$\begin{aligned}
\bar{v} &\leq -\gamma e \leq 0, \\
-\bar{w} &\leq -\xi(\theta + \alpha) \leq \bar{w}.
\end{aligned} \tag{21}$$

Reorganizing the inequalities in (15) (20) and (21), the terminal region can be determined as:

$$\begin{aligned}
0 &\leq \alpha \theta, \\
0 &\leq e \leq -\frac{\bar{v}}{\gamma}, \\
-\frac{\bar{w}}{\xi} &\leq (\theta + \alpha) \leq -\frac{\bar{w}}{\xi}, \\
-\arccos \frac{q_1 + r_1 \gamma^2}{\gamma} &\leq \alpha \leq \arccos \frac{q_1 + r_1 \gamma^2}{\gamma}.
\end{aligned} \tag{22}$$

4. SIMULATION RESULTS

In this section, the simulation results for backward motion of wheeled mobile robots are presented. The cost function is chosen as (6), where the weighting matrix are:

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, R = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}, P = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0.5 & 0.5 \end{bmatrix}.$$

The linear and angular velocities are limited by:

$$\begin{aligned}
-2 &\leq v \leq 0(m/s), \\
-2 &\leq w \leq 2(rad/s).
\end{aligned} \tag{23}$$

The prediction horizon is $T_p = 2.5s$ and sampling interval is $\delta = 0.1s$. Here, coefficients of local control law in (16) are set to be $\gamma = 2$ and $\xi = 2$, respectively. Then the terminal region Ω in (22) (see Fig. 2(a)) are:

$$\begin{aligned}
0 &\leq \alpha \theta, \\
0 &\leq e \leq 1, \\
-1 &\leq (\theta + \alpha) \leq 1, \\
-0.45 &\leq \alpha \leq 0.45.
\end{aligned}$$

For ease of understanding, we represent the terminal region in Cartesian coordinate, as shown in Fig. 2(b).

In order to show the superiority of backward motion algorithm than the forward case in Gu and Hu (2005) when the initial position located in the first and second quadrants, the simulation results are compared and analyzed.

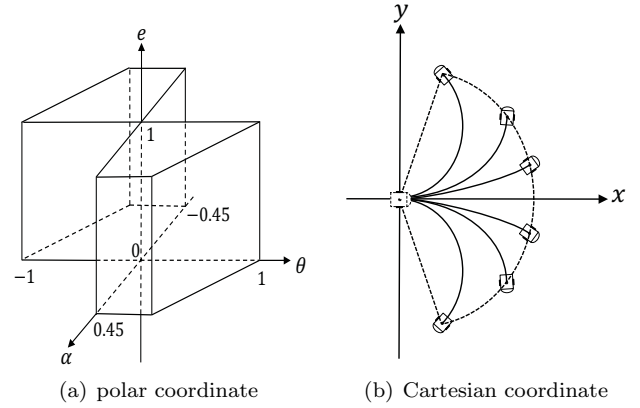


Fig. 2. terminal region Ω defined in (a) polar coordinate (b) Cartesian coordinate

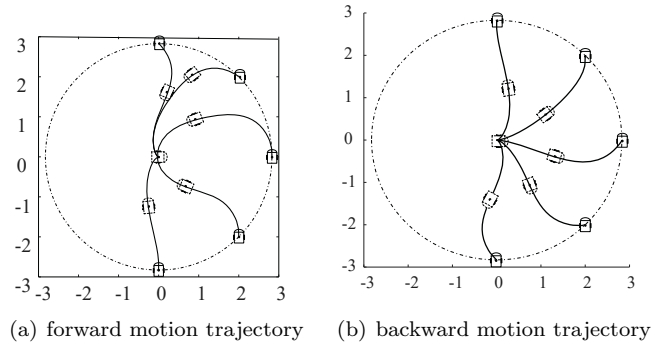


Fig. 3. regulation trajectories.

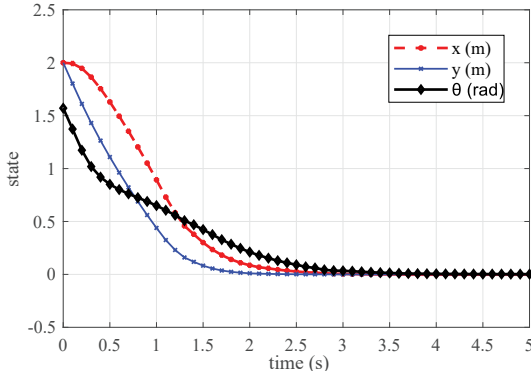
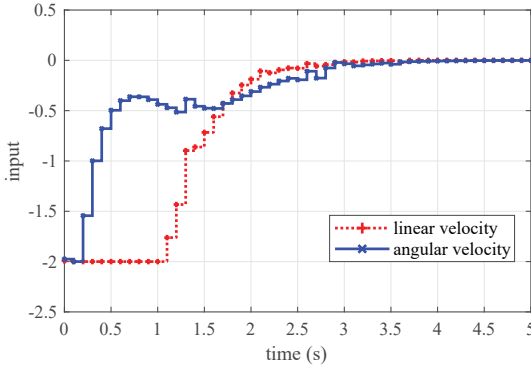
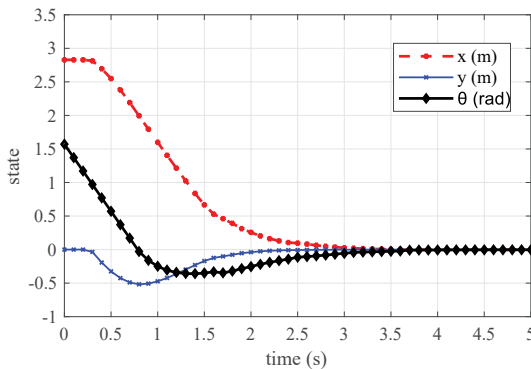
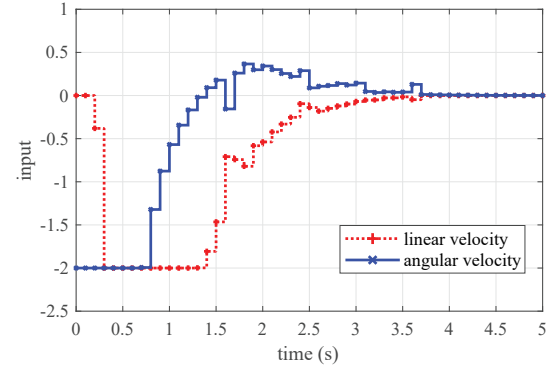
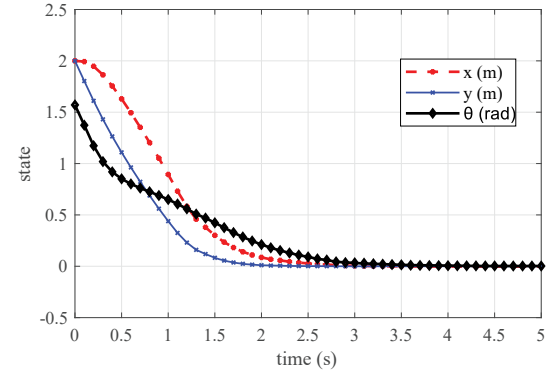
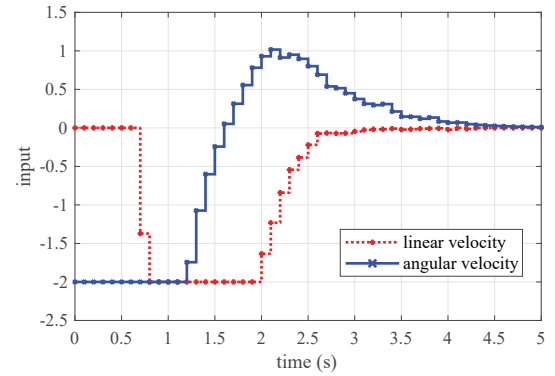
Three different representative initial position are selected as $[x, y, \psi]^T = [2, 2, \pi/2]^T$, $[0, 2\sqrt{2}, \pi/2]^T$, $[2, -2, \pi/2]^T$. Another two initial position are also simulated, which are located in y axis: $[0, 2\sqrt{2}, \pi/2]^T$, $[0, -2\sqrt{2}, \pi/2]^T$. Input constraints are the same, as shown in (23), for the two algorithms.

The trajectory generated by the two algorithms are shown in Fig. 3. Both of them can regulate the wheeled robots to the origin, but the curvature near the origin of backward motion is significantly larger than the forward case. We evaluate the control energy by the value of $\sum_1^n \|\mathbf{u}\|^2$, where n represents the number of iterations before the robot is stabilized. The detailed control energy for different initial positions of the two algorithms are listed in Table 1 and we can conclude that the backward case need less energy than the forward case for the initial positions in the first and second quadrants. But for the initial positions in the negative part of y axis, the result is different, which depends on the initial orientation of the robots.

We use the state variables defined in Cartesian coordinate to show the detailed simulation results for three representative backward trajectories and the corresponding control inputs (see fig.4–9). All the states of three different initial positions converge to zero, which means the proposed algorithm can stabilizing the nonholonomic wheeled mobile robots. The results in Fig.5, Fig.7 and Fig.9 show that the backward algorithm satisfy the input constraints.

Table 1. Control energy for regulation

initial position	forward motion	backward motion
$[0, 2\sqrt{2}, \pi/2]^T$	149.1402	74.4424
$[2, 2, \pi/2]^T$	141.5919	66.8511
$[2\sqrt{2}, 0, \pi/2]^T$	151.8350	88.2080
$[2, -2, \pi/2]^T$	127.1886	115.8113
$[0, -2\sqrt{2}, \pi/2]^T$	74.8921	149.7915

Fig. 4. State trajectory with initial position $[2, 2, \pi/2]^T$ Fig. 5. Input trajectory with initial position $[2, 2, \pi/2]^T$ Fig. 6. State trajectory with initial position $[2\sqrt{2}, 0, \pi/2]^T$ Fig. 7. Input trajectory with initial position $[2\sqrt{2}, 0, \pi/2]^T$ Fig. 8. State trajectory with initial position $[2, -2, \pi/2]^T$ Fig. 9. Input trajectory with initial position $[2, -2, \pi/2]^T$

5. CONCLUSIONS

In this paper, we have designed an MPC algorithm to regulate the nonholonomic wheeled mobile robots for the backward motion case when the initial position located in the first and fourth quadrants. Then the feasibility and stability have been proved, and the terminal region together with local control law have been developed. The simulation results have validated the effectiveness of the proposed backward algorithm. Combining with the forward motion proposed by Gu and Hu (2005), which is more suitable for the initial position located in the second and third quadrants, the two algorithms provide a general framework to efficiently solve the regulation problems of the wheeled mobile robots. In the future, the algorithm

combining the forward and backward motions of the robots under polar coordinate could be further studied.

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